Design to Cost
PART COST ESTIMATOR

2026-10-26

RAW-MATERIAL COST SHEET

EX26-0001 · L mounting bracket

Rev B · Galvanised steel GI · IN · costing month 2026-10 · version 2

Uses example rates – not for negotiation
Replace the EXAMPLE rates marked below on the Rates page first.

Result

	INR	INR
Net RM cost per part	**15.8374**	**15.8374**

Steps and formulas

Step	Formula	Value	Unit
Net weight	volume * density	199.8304	g
Blanks per sheet	best straight grid of blank_length_mm × blank_width_mm on the usable sheet (sheet_length_mm – 2 × edge_mm) × (sheet_width_mm – 2 × edge_mm), gap_mm between blanks, both orientations	189	pcs
Sheet weight (nesting information)	sheet_length_mm * sheet_width_mm * thickness_mm / 1000 * density_g_cm3	49,062	g
Blank weight	blank_length_mm * blank_width_mm * thickness_mm / 1000 * density_g_cm3	205.1676	g
Gross weight (calculated blank + gap)	(blank_length_mm + gap_mm) * (blank_width_mm + gap_mm) * thickness_mm / 1000 * density_g_cm3	234.346	g
Skeleton & punched-out scrap	gross_g - net_weight_g	34.5157	g
Material used per part	gross_g	234.346	g
Material yield	100 * net_weight_g / input_g	85.2715	%
Sheet utilisation	100 * net_weight_g * blanks_per_sheet / sheet_weight_g	76.9792	%
RM cost (material used × rate)	input_g / 1000 * material_rate	16.8729	INR
Scrap credit (–)	0 - returns_g / 1000 * scrap_rate	-1.0355	INR
Net RM cost per part	rm_cost + scrap_credit	15.8374	INR
Net RM cost per part in INR	value * fx	15.8374	INR

Inputs and where each comes from

Input	Value	Unit	Comes from
Volume	25.4561	cm³	3D file example_bracket_revB.step · rev B · Measured from the STEP file with OpenCascade (CadQuery)
Surface area	272.4334	cm²	3D file example_bracket_revB.step · rev B · Measured from the STEP file with OpenCascade (CadQuery)
Projected area (along Z)	97.734	cm²	3D file example_bracket_revB.step · rev B · Measured from the STEP file with OpenCascade (CadQuery)
Density of Galvanised steel GI	7.85	g/cm³	Master data → Materials · Galvanised steel GI
Sheet thickness	2	mm	3D file example_bracket_revB.step · rev B · Most common wall (sheet thickness) found in the 3D model
Flat blank length	326.7	mm	3D file example_bracket_revB.step · rev B · Developed blank from the 3D model: Developed straight strip + bending stage (100 % material yield)
Flat blank width	40	mm	3D file example_bracket_revB.step · rev B · Developed blank from the 3D model: Developed straight strip + bending stage (100 % material yield)
Sheet length	2,500	mm	Master data → Parameters · sheet_metal / sheet_length_mm
Sheet width	1,250	mm	Master data → Parameters · sheet_metal / sheet_width_mm
Gap between blanks	5	mm	Master data → Parameters · sheet_metal / gap_mm
Sheet edge trim	5	mm	Master data → Parameters · sheet_metal / edge_mm
Material rate	72	INR/kg	Rate library #585 · material / Galvanised steel GI · IN · 2026-10 · approved · EXAMPLE
Scrap value	30	INR/kg	Rate library #586 · material_scrap / Galvanised steel GI · IN · 2026-10 · approved · EXAMPLE
INR per INR	1	INR/INR	INR is the base currency.

Rate versions used (the report can be reproduced with these)

Id	Rate	Country	Month	Value	Unit	Status	Approved by	
585	material / Galvanised steel GI	IN	2026-10	72	INR/kg	approved	Example seed	EXAMPLE
586	material_scrap / Galvanised steel GI	IN	2026-10	30	INR/kg	approved	Example seed	EXAMPLE

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